

Strong magnetic fields and contact-interacting few-fermion systems

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My aim is to predict the critical number of particles which form a self-bound state (droplet) while their pair interaction is not attractive enough to sustain a bound dimer. As an intermediate step, I analyze the critical background-field strength for the in-vacuum unstable A -body system to become bound. This field strength shall be analyzed for Bose and Fermi particles.

SCRATCH PAD

$$\begin{aligned}\mathbf{A} &= \frac{1}{2} \mathbf{B} \times \mathbf{r} \\ &= \frac{1}{2} B \rho \hat{\phi}\end{aligned}\tag{1}$$

(for $\mathbf{B} = B\hat{z}$)

with $\rho = \sqrt{x^2 + y^2}$ and the symmetric gauge

$$\mathbf{A} = B \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} . \tag{2}$$

a. Understanding the degeneracy in the spectrum: SVM vs. analytical expectation The wave function of two uncoupled particles trapped in two dimensions with identical harmonic-oscillator potentials is given by

$$\begin{aligned}\phi_{n_{x1}n_{y1}n_{x2}n_{y2}} \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) &= \sqrt{\frac{m\omega}{\hbar}} \cdot \frac{1}{\sqrt{2^{n_{x1}+n_{y1}+n_{x2}+n_{y2}} \cdot \pi \cdot n_{x1}! \cdot n_{y1}! \cdot n_{x2}! \cdot n_{y2}!}} \\ &\cdot e^{-\frac{m\omega}{2\hbar}(x_1^2+y_1^2+x_2^2+y_2^2)} \cdot H_{n_{x1}} \left(\sqrt{\frac{m\omega}{\hbar}} x_1 \right) \cdot H_{n_{y1}} \left(\sqrt{\frac{m\omega}{\hbar}} y_1 \right) \cdot H_{n_{x2}} \left(\sqrt{\frac{m\omega}{\hbar}} x_2 \right) \cdot H_{n_{y2}} \left(\sqrt{\frac{m\omega}{\hbar}} y_2 \right)\end{aligned}\tag{3}$$

with

$$(a_{x1}^\dagger a_{x1} + a_{y1}^\dagger a_{y1} + a_{x2}^\dagger a_{x2} + a_{y2}^\dagger a_{y2}) |n_{x1}n_{y1}n_{x2}n_{y2}\rangle = \hbar\omega (n_{x1} + n_{y1} + n_{x2} + n_{y2} + 2) |n_{x1}n_{y1}n_{x2}n_{y2}\rangle . \tag{4}$$

The expansion

$$\phi = \sum_n c_n g(0; A_n, (x_1, y_1, x_2, y_2)^\top) . \tag{5}$$

cannot approximate $\sum n = \text{odd}$ states because $H_n(x) = (-1)^n H_n(-x)$.

I. INTRODUCTION – WHAT FASCINATES US?

Assume a collection of A non-interacting, non-relativistic identical bosons and imagine you could tune-on a spherically symmetric, attractive pair interaction smoothly. The regime of large enough interaction strength v to form a dimer with zero energy and the associated universal 3-body (original: [? ?]; alternative, potentially generalizable explanation: [? ?]) and cluster spectra (empirical/numerical: [? ? ? ? ? ?]; field-theoretical and classical approach: [? ?])

II. THEORETICAL DESCRIPTION OF STRONGLY INTERACTING FERMIONS IN A MAGNETIC BACKGROUND FIELD

In order to describe the effect of a static magnetic field, which is not necessarily perturbatively small, on the low-energy spectra of small nuclei, the effective-field-theory Hamiltonian of the latter is augmented by, first, minimally coupling the derivatives, and second, by the magnetic-moment 1-body interaction (a remnant of the non-relativistic reduction of the Dirac equation) and the coupling of the $\mathbf{B}$ field to the four-nucleon leading-order momentum-independent vertex.

$$\hat{H} = \tag{6}$$

The so-called Landau spectrum of the minimally coupled, non-interacting limit of this Hamiltonian can be derived analytically

$$E = \tag{7}$$

A. Notable features

1. Total Spin is not conserved, which means that if we consider a 2-nucleon system, it resides in a superposition of spin-1 and spin-0 states.
2. Bound states might not have negative eigenvalues because the non-interacting theory still contains the effect of the magnetic field on the charged particles and shifts their ground state by, at least, the energy in the lowest Landau level.
- 3.

III. TWO-NUCLEON SYSTEMS

First, we analyse the predictions in Ref. [?], i.e, what is the dependence of the lowest eigenvalues of the 2-proton system on the strength of the magnetic field?

IV. THREE NUCLEONS

How does the critical field strength at which the 3-proton system becomes bound compare with the corresponding strength which is necessary to bind two protons?

V. FOUR NUCLEONS

Same critical-strength ratio? Now we extrapolate and predict that no more than $|\mathbf{B}| \sim 4$ T is needed to bind an A -proton cluster.

VI. SVM TECHNICALITIES

Any quantity left undefined is to be found in [?]

$$g(\mathbf{s}; A, \mathbf{x}) = \exp\left\{-\frac{1}{2}\mathbf{x}^\top A \mathbf{x} + \mathbf{s}^\top \mathbf{x}\right\} \tag{generating function}$$

$$\langle g(\mathbf{s}'; A', \mathbf{x}) | \boldsymbol{\pi}^\top \Lambda \boldsymbol{\pi} | g(\mathbf{s}; A, \mathbf{x}) \rangle = \hbar^2 \left(d \operatorname{tr}(AB^{-1}A'\Lambda) - \mathbf{y}^\top \Lambda \mathbf{y} \right) \cdot \underbrace{\left(\frac{(2\pi)^N}{\det B} \right)^{d/2} \cdot \exp\left\{ \frac{1}{2} \mathbf{v}^\top B^{-1} \mathbf{v} \right\}}_{\equiv \langle g(\mathbf{s}'; A', \mathbf{x}) | g(\mathbf{s}; A, \mathbf{x}) \rangle} \quad (E_{\text{kin}} \text{ ME})$$

$$\text{with } B = A + A' \quad , \quad \mathbf{v} = \mathbf{s} + \mathbf{s}' \quad , \quad \mathbf{y} = A'B^{-1}\mathbf{s} - AB^{-1}\mathbf{s}' \quad (8)$$

$$\text{and } A^\top = A \quad , \quad \Lambda^\top = \Lambda \quad , \quad \boldsymbol{\pi}_j = -i\hbar \frac{\partial}{\partial \mathbf{x}_j} \quad (9)$$

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